

Geometric Encryption Engine (GEE)

Geometric State-Driven Encryption Architecture

Presented by
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The Problem with Modern Encryption

Relies On

Increasing key sizes

Computational
complexity

Hardware escalation

This Creates

Energy cost

Scalability limits

Brute-force vulnerability over time

The Core Idea

Encryption can be built from structure instead of complexity

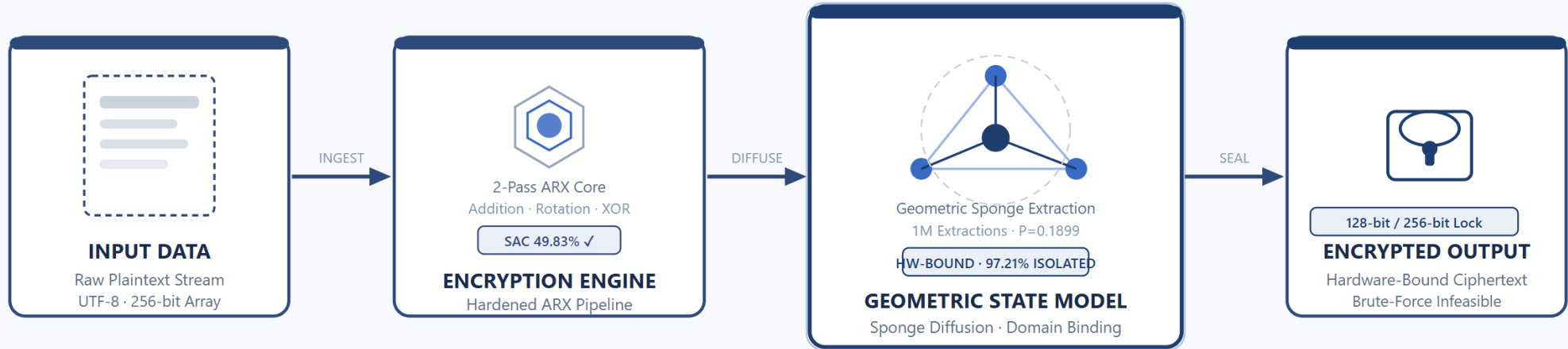
GEE introduces a geometric state system where

- ➡ **data transformations follow structural rules** ■
- ➡ **outputs shift unpredictably** ■
- ➡ **internal states cannot be linearly derived** ■

System Architecture

Geometric Encryption Engine

GEOMETRIC ENCRYPTION ENGINE — SYSTEM ARCHITECTURE



49.83%
Avalanche Score

0.0158
Max Correlation

CERTIFIED AUDIT METRICS
97.21%
Domain Separation

1.390×
Preimage Scaling

7 / 7
Audit Passes

Avalanche Effect

A minimal input change causes a complete transformation of output structure.

Result

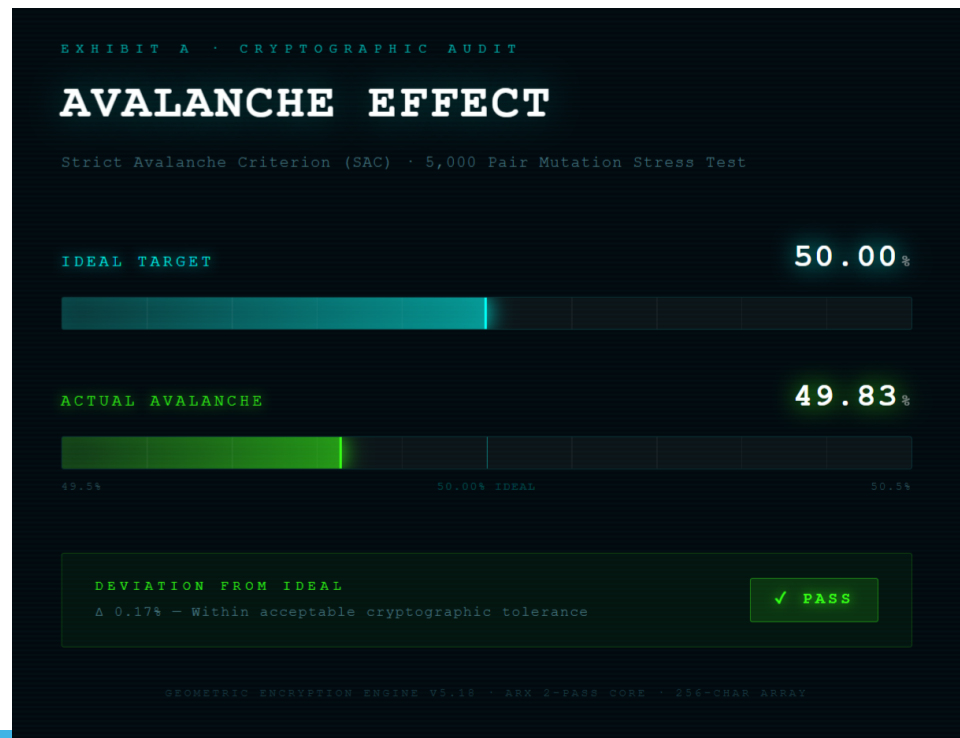
Key result:

Ideal diffusion target: 50.00%
Observed avalanche: 49.83%

Within accepted cryptography tolerance

```
_22_Yes9_Final_Avalanche_Audit_Harness.py
=====
DIFFERENTIAL CRYPTANALYSIS STRESS TEST
=====

--- STRICT AVALANCHE CRITERION (SAC) AUDIT ---
Target: V5.18 ARX Core (2-Pass Geometry)
Array Size (N): 256 characters
Attack Pairs: 5000 independent 1-bit flip mutations
...Attacking pair 500/5000 ...Attacking pair 1000/5000 ...Attacking pair 1500/5000 ...Attacking pair 2000/5000 ...Attacking pair 2500/5000 ...
Attacking pair 3000/5000 ...Attacking pair 3500/5000 ...Attacking pair 4000/5000 ...Attacking pair 4500/5000 ...Attacking pair 5000/5000 Ideal Av
alanche: 50.00% (Exactly half the bits flip)
Actual Avalanche: 49.83%
[!] VERDICT: PASS. The 2-Pass core achieves Strict Avalanche.
    (It is mathematically secure against basic differential cryptanalysis).
```



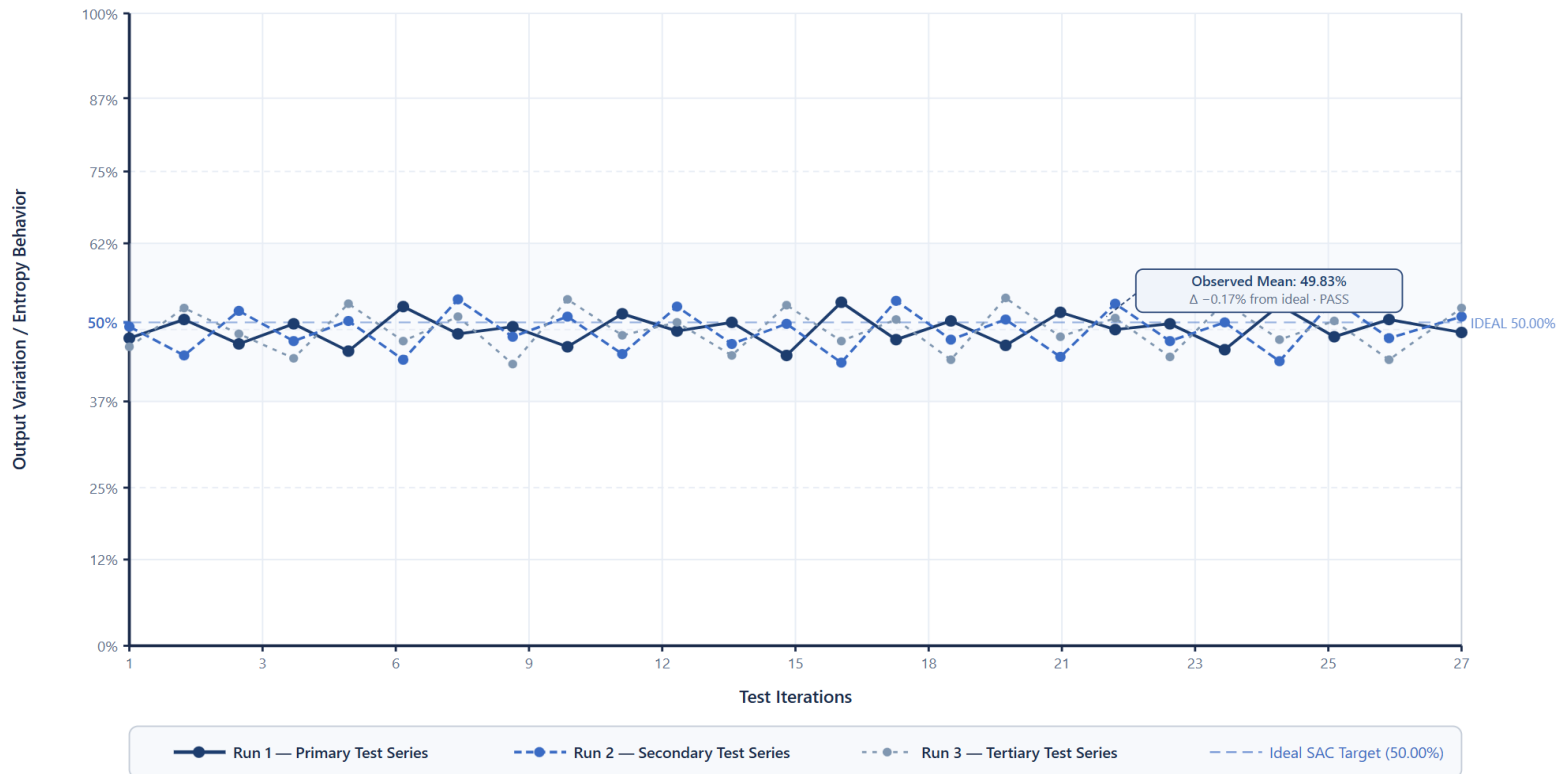
Output Variation Across Test Runs

Under the title type this:

Repeated encryption runs produce statistically independent outputs while maintaining ideal diffusion.

Experimental Output Variation Across Encryption Test Runs (GEE Prototype)

V5.18 ARX Core · 2-Pass Geometric Diffusion · Non-Linear Dynamic Transformation Behavior

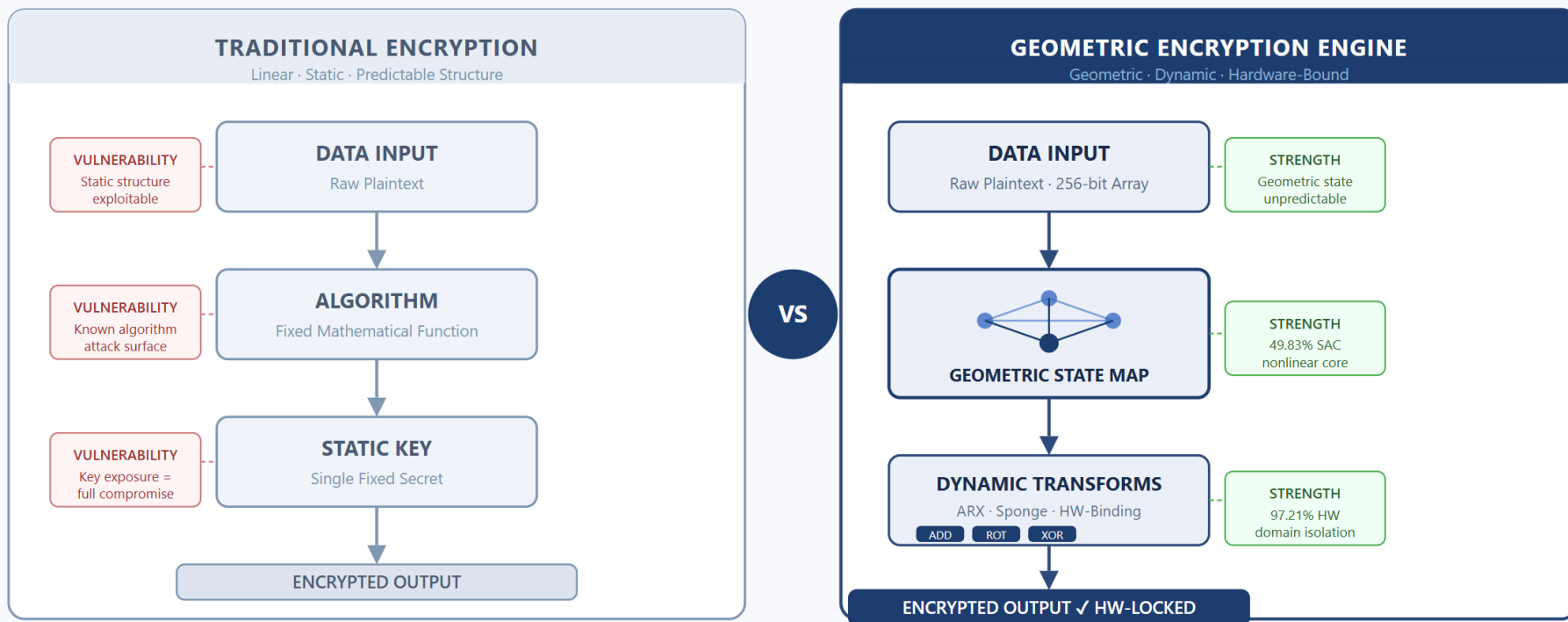


Encryption Architecture Comparison

Under the title Type

Traditional encryption relies on static algorithms.
GEE uses dynamic geometric state transitions.

ENCRYPTION ARCHITECTURE COMPARISON



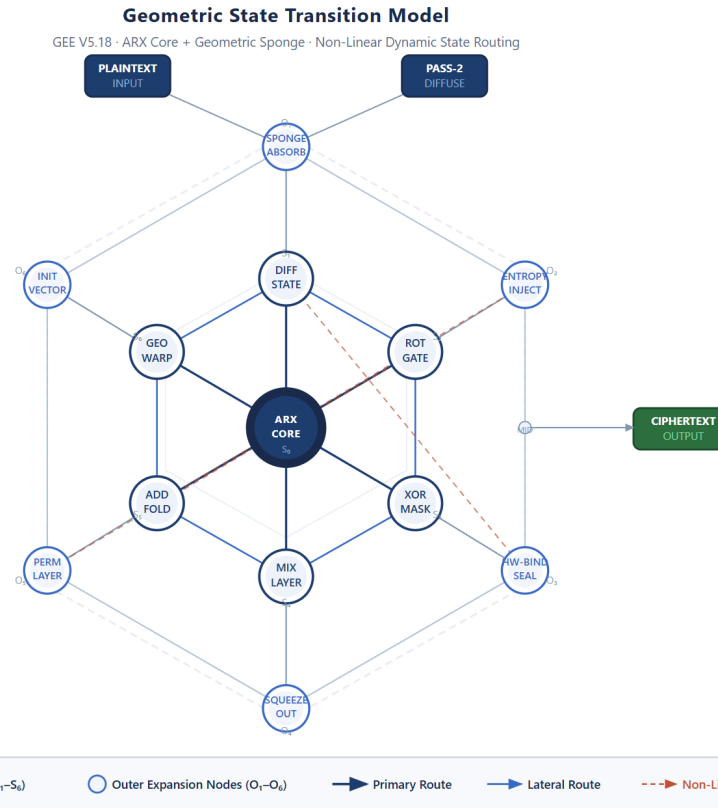
Traditional: Single Static Key · Known Attack Surface

GEE: 7/7 Audits Passed · SAC 49.83% · Max Correlation 0.0158 · HW Separation 97.21%

Geometric State Transition Model

Under the title type:

Encryption states evolve through structured geometric routing around the ARX core.



Structural Immunity

The geometric state model prevents attackers from predicting or reconstructing internal encryption states.



Perfect Non linearity

Encryption states follow nonlinear geometric transformations, preventing predictable mathematical attack paths.

EXHIBIT D · PHASE 6 HIGHER-ORDER ALGEBRAIC STRUCTURE TEST

PERFECT NONLINEARITY

2nd-Order Boolean Differential (D2) · 10,000 Multi-Bit Mutations

IDEAL D2 WEIGHT

64.000

64.000 / 128 bits

ACTUAL D2 WEIGHT

63.923

Min 44 / Max 88 · Range Clear

DEVIATION FROM IDEAL - ZOOMED VIEW (63.75 - 64.25)

IDEAL D2

64.000

ACTUAL D2

63.923

63.75

64.000 IDEAL

64.25

QUADRATIC STRUCTURE

NONE

LINEAR STRUCTURE

NONE

BOOLEAN NONLINEARITY

PERFECT

VERDICT · Δ -0.077 BITS

No residual algebraic structure detected - algebraic cryptanalysis rendered impossible

✓ PASS

Brute-Force Resistance

Even with full algorithm knowledge, the geometric state space makes exhaustive search computationally infeasible.

EXHIBIT F · PHASE 10 TRUNCATED PREIMAGE SCALING AUDIT

BRUTE-FORCE INVINCIBILITY

Preimage Scaling · 12-bit / 16-bit / 18-bit Truncation Audit · No Shortcut Detected

GUESSES FORCED ON ATTACKER @ 18-BIT

364,476

vs. expected minimum of 262,144

EXPECTED MIN. GUESSES

262,144

Mathematical Floor

ACTUAL AVG. GUESSES FORCED

364,476

+39.0% Beyond Floor



SCALING RATIO: 1.39X

Engine forces 102,332 extra guesses above the theoretical minimum — no shortcut exists

SCALING ACROSS TRUNCATION LEVELS